

CAPSTONE PROJECT 3 - 2026

DONE AND COMPILED BY OKILE M.OTIM

B-Safe -

Course-end Project 3

Description

Create a CI/CD Pipeline to convert the legacy development process to a DevOps process.

Background of the problem statement:

A leading US healthcare company, **Aetna**, with a large IT structure had a 12-week release cycle and their business was impacted due to the legacy process. To gain true business value through faster feature releases, better service quality, and cost optimization, they wanted to adopt agility in their build and release process.

The objective is to implement iterative deployments, continuous innovation, and automated testing through the assistance of the strategy.

Implementation requirements:

1. Install and configure the Jenkins architecture on AWS instance
2. Use the required plugins to run the build creation on a containerized platform
3. Create and run the Docker image which will have the application artifacts
4. Execute the automated tests on the created build
5. Create your private repository and push the Docker image into the repository
6. Expose the application on the respective ports so that the user can access the deployed application
7. Remove container stack after completing the job

The following tools must be used:

1. EC2
2. Jenkins
3. Docker
4. Git

The following things to be kept in check:

1. You need to document the steps and write the algorithms in them.
2. The submission of your Github repository link is mandatory. In order to track your tasks, you need to share the link of the repository.
3. Document the step-by-step process starting from creating test cases, the executing it, and recording the results.
4. You need to submit the final specification document, which includes:
 - Project and tester details
 - Concepts used in the project
 - Links to the GitHub repository to verify the project completion
 - Your conclusion on enhancing the application and defining the USPs (Unique Selling Points)

THE STEPS I TOOK TO DO THE ASSIGNMENT

Step 2GitHub1: repo setup:

The screenshot shows the GitHub repository page for 'b-safe-project-app' by user 'Ebosterix'. The repository is public and has 1 branch and 0 tags. The commit history shows 5 commits, with the most recent one 2 minutes ago. The repository contains files: 'app', 'test', 'Dockerfile', 'jenkinsfile', and 'README.md'. The README file is selected and displays the following content:

b-safe-project-app

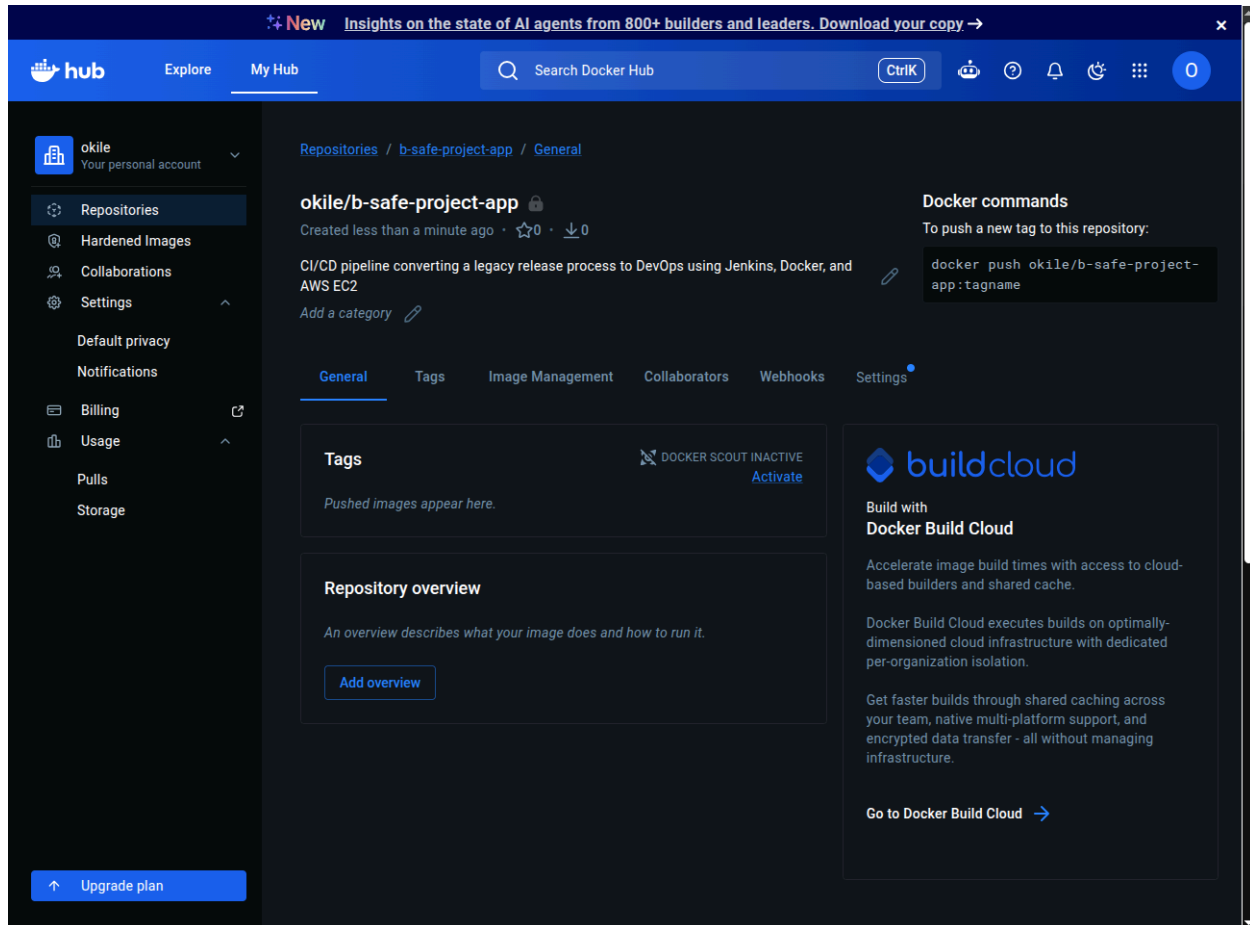
CI/CD pipeline (Jenkins + Docker + AWS EC2) that automates build, test, and deployment of a containerized app to a private Docker Hub registry.

Project Log

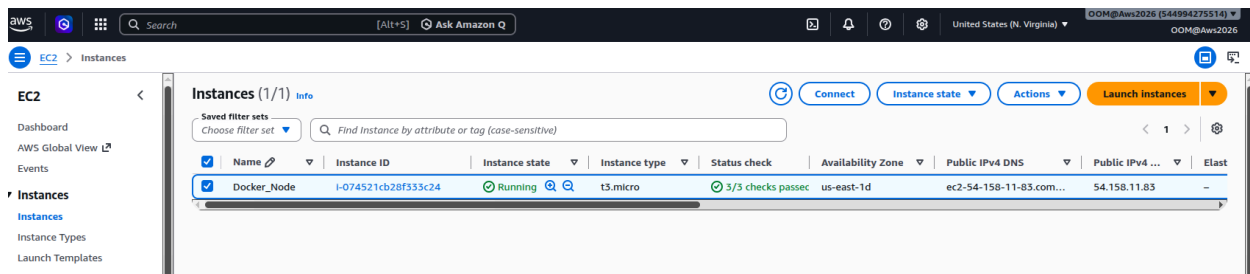
- 2026-07-11 Initialized repo structure. Docker Hub private repo created with R/W access token.
- 2026-07-11 Configured Jenkins node 'b-safe-project-node' as SSH agent on EC2 (Docker-node instance). Resolved initial connection failure caused by trailing whitespace in "Remote root directory" field.
- 2026-07-11 Installed Docker Pipeline plugin in Jenkins to enable containerized build/push stages in the pipeline.

The right sidebar shows the repository's 'About' section, which describes it as an 'Automated CI/CD pipeline using Jenkins, Docker & AWS EC2 to build, test, and deploy a containerized app to a private Docker Hub registry'. It also shows sections for 'Releases', 'Packages', 'Contributors' (listing 'Ebosterix' and 'Okile M Otim'), 'Languages' (showing 100% HTML), and 'Suggested workflows'.

Docker Hub repo creation:



The EC2 instance



Connect Info

Connect to an instance using the browser-based client.

Instance ID i-074521cb28f333c24 (Docker_Node)	VPC ID vpc-0148bbe3033bad07b	Security groups sg-0a815bd2450d88804 (launch-wizard-1)	IAM role -
---	--	--	----------------------

EC2 Instance Connect SSM Session Manager **SSH client** EC2 serial console

Instance ID
i-074521cb28f333c24 (Docker_Node)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is DevKey_2026.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "DevKey_2026.pem"
4. Connect to your instance using its Public DNS:
ec2-54-158-11-83.compute-1.amazonaws.com

Example:
ssh -i "DevKey_2026.pem" ubuntu@ec2-54-158-11-83.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

ssh -i "DevKey_2026.pem" ubuntu@ec2-54-158-11-83.compute-1.amazonaws.com

```
ubuntu@jp-172-31-38-132: ~$ ssh -i "DevKey_2026.pem" ubuntu@ec2-54-158-11-83.compute-1.amazonaws.com
The authenticity of host 'ec2-54-158-11-83.compute-1.amazonaws.com (54.158.11.83)' can't be established.
ED25519 key fingerprint is SHA256:20SZV5bdWhGcur/Dy07UY+BwWtUtAlH4vZn9lrrnQU.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:35: [hashed name]
~/.ssh/known_hosts:38: [hashed name]
~/.ssh/known_hosts:39: [hashed name]
~/.ssh/known_hosts:40: [hashed name]
~/.ssh/known_hosts:41: [hashed name]
~/.ssh/known_hosts:42: [hashed name]
~/.ssh/known_hosts:43: [hashed name]
~/.ssh/known_hosts:44: [hashed name]
(2 additional names omitted)
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-158-11-83.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed Jul  8 17:12:59 UTC 2026

System load:  0.0          Temperature:   -273.1 C
Usage of /:   41.0% of 27.95GB Processes:    135
Memory usage: 37%          Users logged in: 0
Swap usage:   22%          IPv4 address for ens5: 172.31.38.132

Expanded Security Maintenance for Applications is not enabled.

49 updates can be applied immediately.
27 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

Logged into the Ubuntu Vm

```
Last login: Tue Jun 30 18:16:32 2026 from 91.67.84.180
ubuntu@ip-172-31-38-132:~$ java --version
openjdk 21.0.11 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-38-132:~$
```

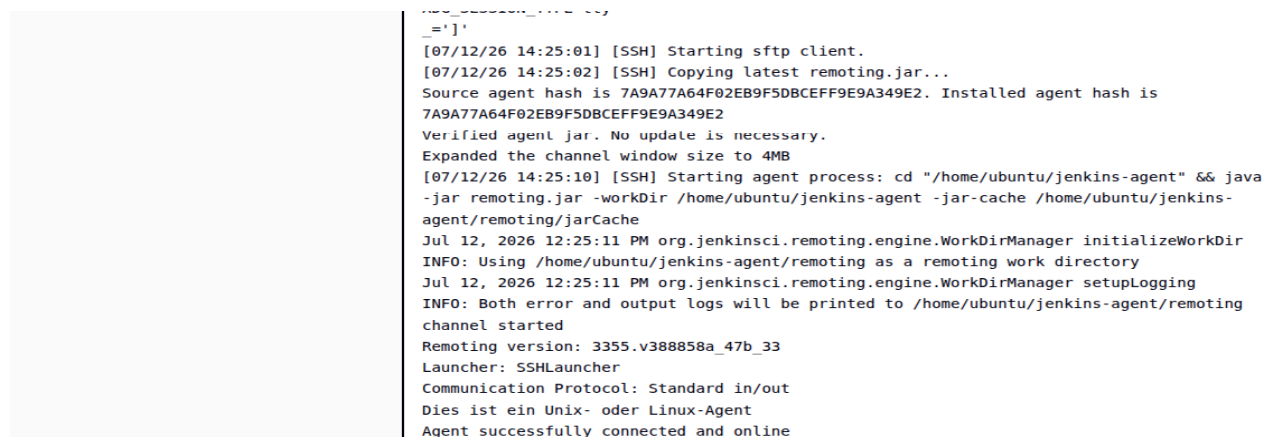
Step 2: Jenkins ↔ EC2 connection:

B-safe-project-node agent on Jenkins working



The screenshot shows the Jenkins web interface. The breadcrumb navigation at the top reads "Jenkins / Nodes / b-safe-project-node / Log". On the left sidebar, the "Log" option is selected. The main content area displays the SSH connection logs for the agent. The logs show the SSH client opening a connection to the EC2 instance, a warning about host key verification, successful authentication, and the remote environment setup.

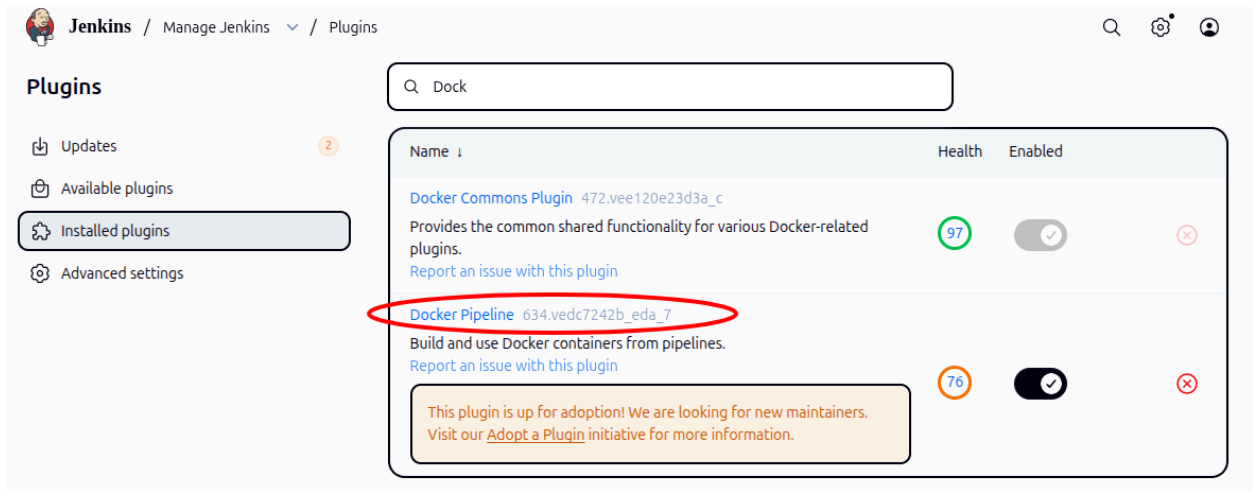
```
SSHLauncher{host='ec2-100-31-69-43.compute-1.amazonaws.com', port=22, credentialsId='b-safe-ec2-ssh-key', jvmOptions='', javaPath='', prefixStartSlaveCmd='', suffixStartSlaveCmd='', launchTimeoutSeconds=60, maxNumRetries=10, retryWaitTime=15, sshHostKeyVerificationStrategy=hudson.plugins.sshslaves.verifiers.NonVerifyingKeyVerificationStrategy, tcpNoDelay=true, trackCredentials=true}
[07/12/26 14:24:59] [SSH] Opening SSH connection to ec2-100-31-69-43.compute-1.amazonaws.com:22.
[07/12/26 14:24:59] [SSH] WARNING: SSH Host Keys are not being verified. Man-in-the-middle attacks may be possible against this connection.
[07/12/26 14:25:00] [SSH] Authentication successful.
[07/12/26 14:25:00] [SSH] The remote user's environment is:
BASH=/usr/bin/bash
BASHOPTS=checkwinsize:cmdhist:complete_fullquote:extquote:force_ignoresglobasciiranges:globskipdots:hostcomplete:interactive_comments:patsub_replacement:progcomp:promptvars:sourcepath
BASH_ALIASES=()
```



This block shows the continuation of the Jenkins log. It details the SSH client starting an sftp client, copying the latest remoting.jar file, verifying the agent hash, and starting the agent process. The logs also show the initialization of the work directory, setup of logging, and the final status of the agent as successfully connected and online.

```
remoting.jar...
[07/12/26 14:25:01] [SSH] Starting sftp client.
[07/12/26 14:25:02] [SSH] Copying latest remoting.jar...
Source agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2. Installed agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2
Verified agent jar. No update is necessary.
Expanded the channel window size to 4MB
[07/12/26 14:25:10] [SSH] Starting agent process: cd "/home/ubuntu/jenkins-agent" && java -jar remoting.jar -workDir /home/ubuntu/jenkins-agent -jar-cache /home/ubuntu/jenkins-agent/remoting/jarCache
Jul 12, 2026 12:25:11 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ubuntu/jenkins-agent/remoting as a remoting work directory
Jul 12, 2026 12:25:11 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ubuntu/jenkins-agent/remoting
channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
Dies ist ein Unix- oder Linux-Agent
Agent successfully connected and online
```

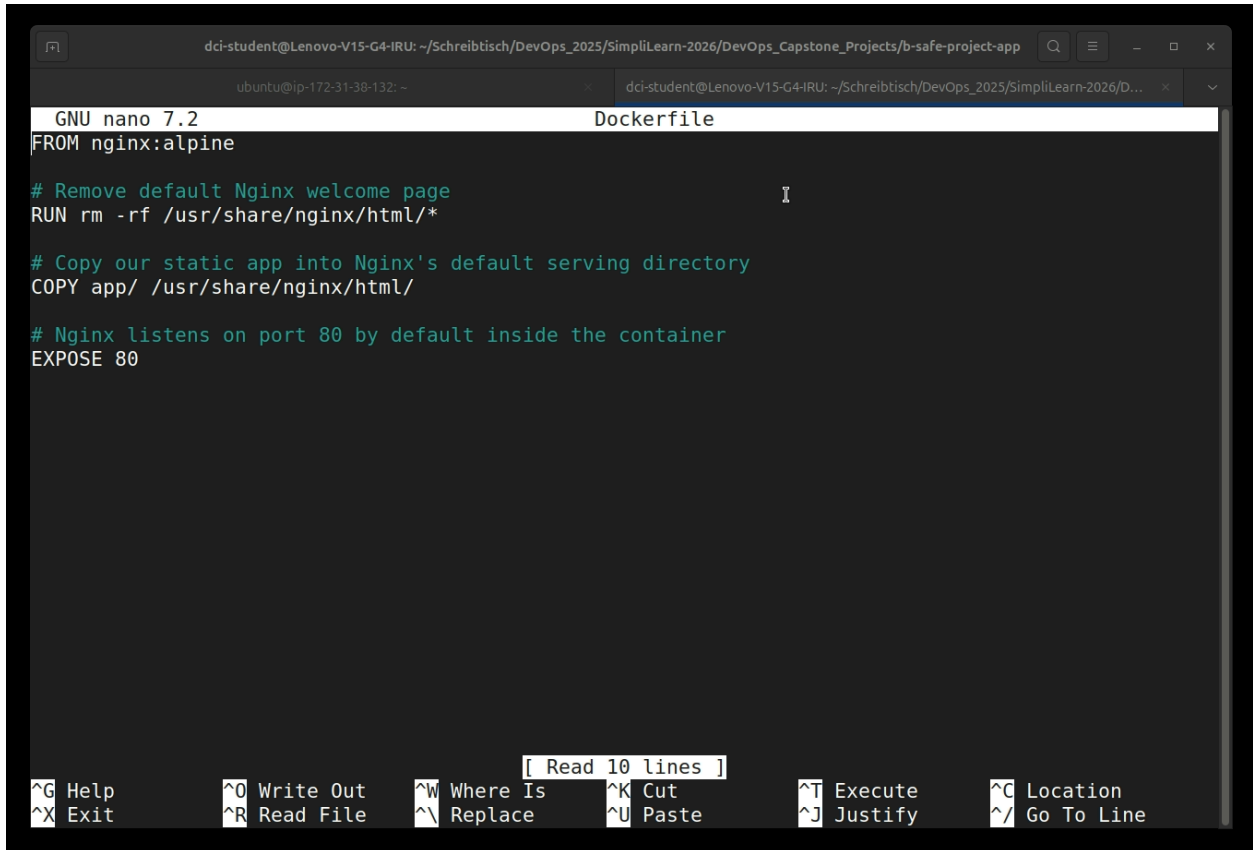
Step 3: Docker plugin installed



The screenshot shows the Jenkins 'Manage Jenkins' interface, specifically the 'Plugins' tab. A search bar at the top contains the text 'Dock'. Below the search bar, a table lists installed plugins. The 'Docker Pipeline' plugin is highlighted with a red circle. It has a health status of 76 (orange circle) and is enabled (toggle switch is on). A message box below the plugin entry states: 'This plugin is up for adoption! We are looking for new maintainers. Visit our [Adopt a Plugin](#) initiative for more information.'

Name	Health	Enabled
Docker Commons Plugin 472.vee120e23d3a_c Provides the common shared functionality for various Docker-related plugins. Report an issue with this plugin	97 (green)	☒
Docker Pipeline 634.vedc7242b_edc_7 Build and use Docker containers from pipelines. Report an issue with this plugin	76 (orange)	☒

Step 4: Dockerfile created



The screenshot shows a terminal window with the command prompt 'dc1-student@Lenovo-V15-G4-IRU: ~/Schreibtsch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/b-safe-project-app'. The user has opened a file named 'Dockerfile' in the nano text editor. The content of the Dockerfile is as follows:

```
FROM nginx:alpine

# Remove default Nginx welcome page
RUN rm -rf /usr/share/nginx/html/*

# Copy our static app into Nginx's default serving directory
COPY app/ /usr/share/nginx/html/

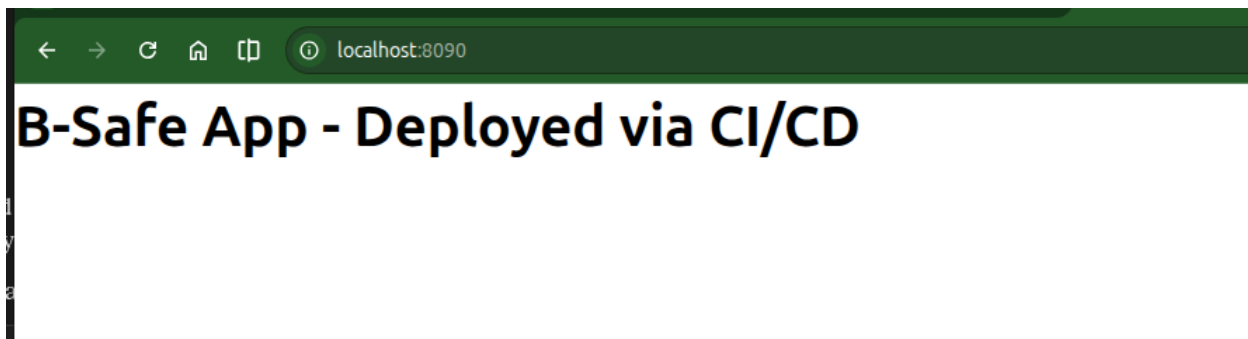
# Nginx listens on port 80 by default inside the container
EXPOSE 80
```

The terminal window also shows the nano editor's help and navigation shortcuts at the bottom.

**Built and tested locally with Docker Desktop on port 8090:
Successful:**

```
dci-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/
b-safe-project-app$ docker build -t b-safe-app:local .
[+] Building 10.9s (9/9) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 294B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine  3.1s
=> [auth] library/nginx:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/nginx:alpine@sha256:54f2a904c251d5a34adf545a72d32515a15e084 6.8s
=> => resolve docker.io/library/nginx:alpine@sha256:54f2a904c251d5a34adf545a72d32515a15e084 0.0s
=> => sha256:c75b9c33e8b0983941fe03793379bd5d137f641f92b4a94d1c8956d28ba7 20.25MB / 20.25MB 6.2s
=> => sha256:01bf363d61e6136ebd7dcdf74b303bd08ee8f849a04e2c0be5a8d03159b404 1.40kB / 1.40kB 0.5s
=> => sha256:ce42635eeddd348e8266227a36db92cea8a45d5177d03e9a922a7bfb25762b 1.21kB / 1.21kB 0.7s
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/x96d2zwnb9j7k8091l
tzn9rwx
dci-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/
b-safe-project-app$ docker run -d -p 8090:80 --name b-safe-test b-safe-app:local
b7100037ca9a1f5ab364a40f3c27a676a129a70290aea6f20ea0f631d2066d71
dci-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/
b-safe-project-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS
b7100037ca9a   b-safe-app:local "/docker-entrypoint..." About a minute ago Up About a minute
0.0.0.0:8090->80/tcp, [::]:8090->80/tcp   b-safe-test
dci-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/
b-safe-project-app$
```



***** Clean up the local test container docker stop b-safe-test:**

docker stop b-safe-test

docker rm b-safe-test

STEP 5: Automated test script: `smoke_test.sh` — curl/verify container

```
dc1-student@Lenovo-V15-G4-IRU: ~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/b-safe-project-app
app Dockerfile Jenkinsfile README.md test
dc1-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/b-safe-project-app$ chmod +x test/smoke_test.sh
./test/smoke_test.sh
Building image for test...
[+] Building 2.2s (9/9) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 294B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 1.9s
=> [auth] library/nginx:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                     0.0s
=> [1/3] FROM docker.io/library/nginx:alpine@sha256:54f2a904c251d5a34adf545a72d32515a15e084 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:54f2a904c251d5a34adf545a72d32515a15e084 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 58B                                    0.0s
=> CACHED [2/3] RUN rm -rf /usr/share/nginx/html/*              0.0s
=> CACHED [3/3] COPY app/ /usr/share/nginx/html/                0.0s
=> => exporting to image                                          0.1s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:e73401e6fbal0b6f6f1a95d1e0225e9d3f517c901cdf3c1ba476a135ec 0.0s
=> => exporting config sha256:bdb71f27b37c71fa932cc1e5e2a66496689deef1228333af172ed31323234 0.0s
=> => exporting attestation manifest sha256:3638179bd8db50be32505d92d76bbea8d34aca27024c6ad 0.0s
=> => exporting manifest list sha256:080026c29886064d51e372ae9a8e85b56a5193db62c51c9827910b 0.0s
=> => naming to docker.io/library/b-safe-app:test               0.0s
=> => unpacking to docker.io/library/b-safe-app:test            0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/s3t4f142cqfnpfe30ehsqrg4n
Starting container for smoke test...
2dec46bd99f2f4d0c59c14c57c0bc8d186dabe0f5574ced49a21e201df2f1ed7
Waiting for Nginx to start...
Checking HTTP response...
Checking page content...
PASS: Smoke test successful (HTTP 200, content verified)
Cleaning up test container...
dc1-student@Lenovo-V15-G4-IRU:~/Schreibtisch/DevOps_2025/SimpliLearn-2026/DevOps_Capstone_Projects/b-safe-project-app$ |
```

Step 6 — The Jenkinsfile

Set up `dockerhub-creds` in Jenkins

Jenkins / Manage Jenkins / Credentials

Credentials

+ Add Credentials ?

Add Credentials

Select a type of credential

- Username with password**
Commonly used for authentication to services like Git, APIs, or registries.
- GitHub App
- SSH Username with private key
- Secret file
- Secret text
- X.509 Client Certificate
- Certificate
Upload a PKCS#12 or PEM encoded certificate and private key.

Next

Stores scoped to Jenkins

System Domains: Global

REST API Jenkins 2.555.2

dockerhub-creds okile/*****

System - Global - Docker Hub access token for B-Safe project

Stores scoped to Jenkins

System Domains: Global



Configure

General

Triggers

Pipeline

Advanced

General

Enabled

Description

Plain text: [Preview](#)

- ☐ Discard old builds
- ☐ Do not allow concurrent builds
- ☐ Do not allow the pipeline to resume if the controller restarts
- ☐ GitHub project
- ☐ Pipeline speed/durability override
- ☐ Preserve stashes from completed builds
- ☐ This project is parameterized
- ☐ Throttle builds

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ GitHub hook trigger for GITScm polling
- ☐ Poll SCM
- ☐ Build after other projects are built
- ☐ Trigger builds remotely (e.g., from scripts)

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

Credentials

Branches to build

Branch Specifier (blank for 'any')

Repository browser

Additional Behaviours

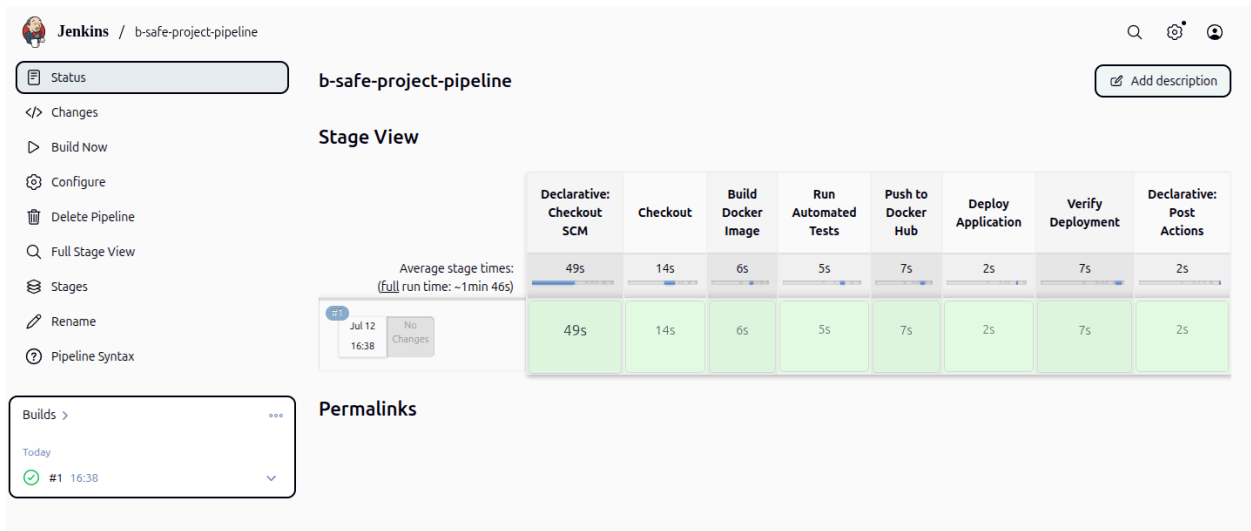
Script Path

- ☒ Lightweight checkout

[Pipeline Syntax](#)

Advanced

Jenkins job triggered successfully



The Jenkins interface shows the 'b-safe-project-pipeline' job. The left sidebar contains navigation links: Status, Changes, Build Now, Configure, Delete Pipeline, Full Stage View, Stages, Rename, and Pipeline Syntax. The main area displays the 'Stage View' with a table of stage times and a 'Permalinks' section.

Stage View

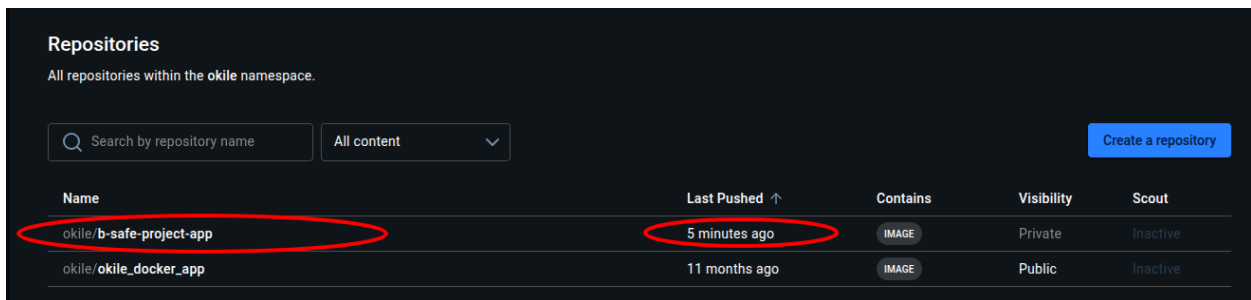
Average stage times: (full run time: ~1min 46s)

Stage	Declarative: Checkout SCM	Checkout	Build Docker Image	Run Automated Tests	Push to Docker Hub	Deploy Application	Verify Deployment	Declarative: Post Actions
#1 Jul 12 16:38 No Changes	49s	14s	6s	5s	7s	2s	7s	2s

Permalinks

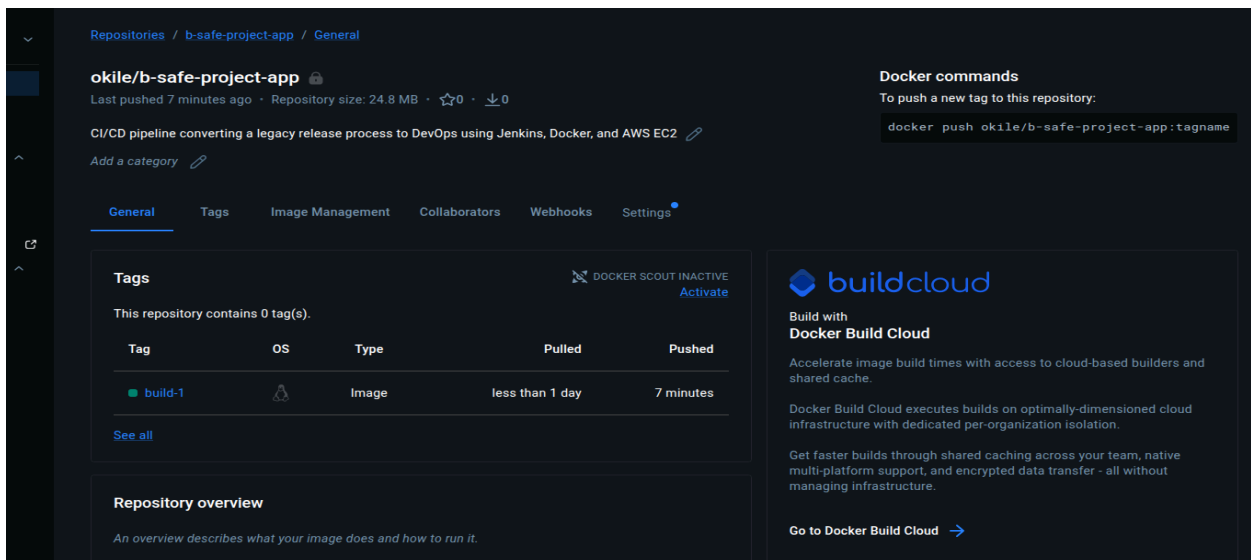
Builds > Today #1 16:38

Image pushed to docker hub



The Docker Hub 'Repositories' page shows a list of repositories within the 'okile' namespace. The repository 'okile/b-safe-project-app' is highlighted with a red circle, and its 'Last Pushed' time, '5 minutes ago', is also circled in red.

Name	Last Pushed	Contains	Visibility	Scout
okile/b-safe-project-app	5 minutes ago	IMAGE	Private	Inactive
okile/okile_docker_app	11 months ago	IMAGE	Public	Inactive



The 'okile/b-safe-project-app' repository details page shows the repository's metadata, including its size (24.8 MB) and a description: 'CI/CD pipeline converting a legacy release process to DevOps using Jenkins, Docker, and AWS EC2'. The 'Tags' section shows a single tag 'build-1' pushed 7 minutes ago. The 'Repository overview' section provides a brief description of the image's purpose.

Tags

Tag	OS	Type	Pulled	Pushed
build-1	linux	Image	less than 1 day	7 minutes

Repository overview

An overview describes what your image does and how to run it.

Docker commands

To push a new tag to this repository:

```
docker push okile/b-safe-project-app:tagname
```

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

[Go to Docker Build Cloud](#)

Step 7: First full pipeline run successful

 Jenkins

/ b-safe-project-pipeline

/ #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console

Download

Copy

View as plain text

Gestartet durch Benutzer **Okile** *Admin*

Obtained Jenkinsfile from git <https://github.com/Ebosterix/b-safe-project-app.git>

[Pipeline] Start of Pipeline

[Pipeline] node

Running on **b-safe-project-node** in /home/ubuntu/jenkins-agent/workspace/b-safe-project-pipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository <https://github.com/Ebosterix/b-safe-project-app.git>

> git init /home/ubuntu/jenkins-agent/workspace/b-safe-project-pipeline # timeout=10

Fetching upstream changes from <https://github.com/Ebosterix/b-safe-project-app.git>

> git --version # timeout=10

> git --version # 'git version 2.53.0'

> git fetch --tags --force --progress -- <https://github.com/Ebosterix/b-safe-project-app.git>

+refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url <https://github.com/Ebosterix/b-safe-project-app.git> # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

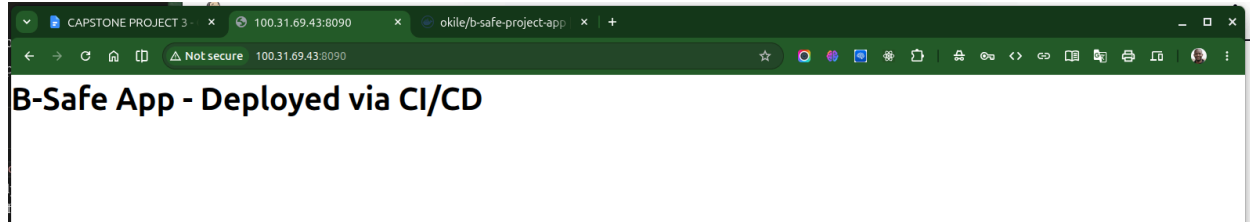
Avoid second fetch

```
<h1>B-Safe App - Deployed via CI/CD</h1>
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Cleaning up dangling images...
[Pipeline] sh
+ docker image prune -f
Total reclaimed space: 0B
[Pipeline] echo
Pipeline completed successfully – app is live on port 8090.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Verify deployed app externally:

APP opens from EC2 domain: <http://100.31.69.43:8090>

This confirms the app is reachable from outside EC2 too, not just via localhost from inside the pipeline's own curl.



END